# Assessing Video LLM Performance in Detecting Child Safety Risks

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## INTRODUCTION

- Over 4 million children are treated for unintentional household injury per year [2], with many likely due to the limits of human attention.
- Traditional child monitoring devices possess critical limitations, namely:
  1) They require constant and scrutinous supervision.,
  2) They are restricted to static, pre-defined infant behaviors.
  3) They act as passive rather than proactive alert systems.
- Increases in remote work, childcare costs, and the adoption of smart home technology suggests the existence of a demand for improved child safety tools.
- Advancements in multimodal large language models has yielded enhanced video understanding, behavior recognition and complex reasoning capabilities.

## OBJECTIVE

- Evaluate two multimodal LLMs for their efficacy in detecting child safety risks.
- Determine the feasibility of developing an AI-enabled child safety monitor that matches or enhances human perception.

## METHODS

### Models Evaluated

The study tested two vision-language models on T4 GPUs:

- **Qwen2-VL-2B-Instruct** [3] (using INT8 quantization)
- **Video-LLaVa-7B** [1] (using 4-bit quantization)

### Data Collection

Both models were passed **45 use-case videos** of children engaging in either dangerous (e.g., climbing unstable surfaces or nearing electrical hazards) or age-appropriate activity (e.g., playing with toys or eating whilst supervised).

### Video Analysis

Videos were processed using **overlapping 5-second windows with a 2-second stride** to **simulate live video streaming**. For each window, 4-8 frames were sampled and compiled into a temporary video clip for analysis.

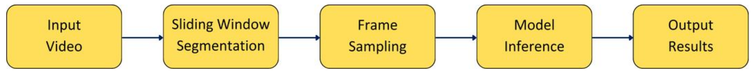



**Figure 1. Diagram Illustrating the Video Analysis Approach**

## METHODS

Each clip was analyzed using the following prompt:

> **Context**: "You are a child safety monitor. Assess this situation from a CHILD SAFETY perspective and evaluate if the child is engaging in unsafe behavior right now based on what you observe. Focus on: actively dangerous actions/behaviors, potential hazards, or immediate threats that will likely cause injury, especially if unsupervised."

> **Question**: "Is this child in any immediate danger? Consider what specific risks you see and why this is dangerous for a CHILD specifically and then begin your response with either 'Yes, the child is in danger' or 'No, the child is not in danger'."

**Output**

- Full model responses to prompt
- An annotated video with danger overlays
- **Record of the first instance of danger detected** (in both seconds and frame number)

**Evaluation Metrics**

The models' performance was compared against researcher perception at both **temporal** (second-level) and **granular** (frame-level) precision.

## RESULTS

As shown in Figures 2 and 3:

- On **31/45 videos** evaluated at second-level granularity and **29/45 videos** evaluated at frame-level precision, Qwen2-VL's predictions of the first instance of danger detected most closely matched the human perceptual baseline.
- On **23/45 videos** evaluated at both temporal and frame-level granularity, Video-LLaVa's predictions most closely matched the human perceptual baseline.

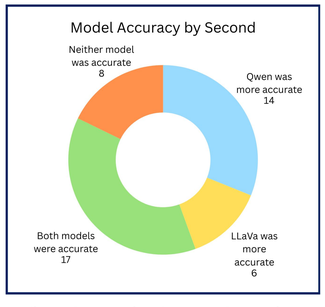



**Figure 2. Detection Accuracy Comparison Measured in Seconds**

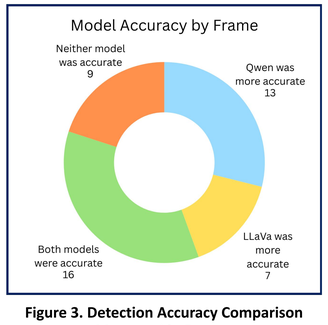



**Figure 3. Detection Accuracy Comparison Measured in Frames**

## RESULTS

- Video-LLaVa outperformed Qwen2-VL in only 6 cases.
- Both models successfully identified danger simultaneously with humans in 17/45 instances.

Overall, as shown in Figure 4, Qwen2-VL demonstrated a **67% average accuracy rate**, while Video-LLaVa achieved a **lower accuracy rate of 51%**.

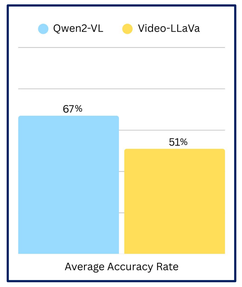



**Figure 4. Average Detection Accuracy Percentage of Both Models**

## CONCLUSION

The results of this preliminary study demonstrate strong potential for the use of multimodal AI for automated child danger detection, bridging the gap left by traditional monitoring devices. Qwen2-VL's performance in handling diverse and dangerous scenarios suggests its superiority for this application.

**Limitations**

- Lack of temporal context
- Difficulty with nuanced risks and judgement
- Single human rater

**Future Work**

- Fine-tune the models and newer models.
- Supply context from previous video windows to improve situational awareness
- Collect additional data to refine precision for eventual deployment

## REFERENCES

**[1]** Lin, B., Ye, Y., Zhu, B., Cui, J., Ning, M., Jin, P., & Yuan, L. (2023). Video-LLaVA: Learning United Visual Representation by Alignment Before Projection. arXiv. https://doi.org/10.48550/ARXIV.2311.10122

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**[3]** Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., Fan, Y., Dang, K., Du, M., Ren, X., Men, R., Liu, D., Zhou, C., Zhou, J., & Lin, J. (2024). Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution. arXiv. https://doi.org/10.48550/ARXIV.2409.12191

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